International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0383

First Published Online: Dec. 29, 2023

Human Kinetics
ORIGINAL INVESTIGATION

Postactivation Performance Enhancement With Maximal Isometric Contraction on Power-Clean Performance Across Multiple Sets

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Purpose*:** This study investigated the postactivation performance-enhancement effect of maximal voluntary isometric contraction (MVIC) at the starting position on power-clean performance over a series of contrast sets. ***Methods*:** Eighteen male (age: 31 [3.7] y, body mass: 76.8 [9.1] kg, height: 175.0 [5.2] cm) and 2 female (age: 27.5 [3.5] y, body mass: 53.3.8 [2.0] kg, height: 158.5 [4.9] cm) resistance-trained individuals performed a contrast postactivation performance-enhancement protocol (isometric contrast training condition [ISO]) consisting of 3 sets of 3 MVICs alternated with 3 power cleans, with an intracontrast rest period of 1 minute. A control protocol consisted of 3 sets of 3 power cleans were performed in a separate session. Barbell velocity during the power clean was measured as an indicator of performance. ***Results: A significant time effect was observed for both mean velocity (MV; $P < .001$) and peak velocity (PV; $P = .008$). Time × group ($P = .415–.444$) and group ($P = .158–.210$) effects showed no significant difference for either MV or PV. However, differences in MV and PV between the corresponding sets of ISO and control condition exceeded the minimum worthwhile change, showing a small to moderate effect (MV: $d = 0.38–0.50$, PV: $d = 0.35–0.50$) in favor of ISO. There was no significant difference in rating of perceived exertion between conditions ($P = .385$, $d = 0.22$). ***Conclusion:*** Power-clean performance was potentiated after 1 minute of rest following 3 repetitions of MVIC across 3 sets. Furthermore, the ISO protocol did not result in greater perception of exertion. These results indicate that coaches may incorporate MVICs as the postactivation performance-enhancement stimulus during contrast training involving the power-clean exercise.

***Keywords*:** contrast training, heavy resistance exercise, warm-up, barbell velocity

Postactivation performance enhancement (PAPE) is a phenomenon where increases in voluntary muscle contraction occur following high-intensity activity, leading to improved sport-related skill performance.[1–3] PAPE can be attained by various conditioning activities including lifting heavy weights, performing plyometric and ballistic exercises with variable resistance, and performing maximal voluntary isometric contraction (MVIC).[2–4] These activities are usually performed with high intensity and low volume to maximize potentiation while minimizing fatigue as high fatigue level may negate any potentiation effect of the conditioning activities.[1–3] In view of this, performing MVIC to induce PAPE may be more advantageous than heavy resistance exercise as it has been reported that individuals may recover faster from isometric than heavy resistance exercises.[5] The use of MVIC of lower-limb muscles have been reported to improve countermovement jump, squat jump, drop jump, and sprint performance in several studies.[4,6–10] However, Robbins and Docherty[11] reported no change in countermovement jump performance after 3 × 7-second isometric back squat at MVIC. A possible reason for this discrepancy could be the duration of contraction as the 3 × 7-second contraction by Robbins and Docherty[11] might have resulted in increased level of fatigue as compared with studies that used a protocol with shorter contraction duration (≤5 s per repetition and ≤15 s for total contraction volume).[6,8,10,12,13]

Despite the PAPE effect of MVIC on jump performance, there are no studies conducted to investigate the effect of MVIC on ballistic heavy resistance lifting performance (ie, velocity or power when lifting a given load or heaviest load lifted), such as weightlifting and weightlifting derivatives. Such exercises are often used in strength training to improve lower-limb force generation capabilities.[14,15] Hence, the need for a study to investigate the PAPE effect of MVIC on ballistic heavy resistance lifting performance.

Importantly, isometric force produced at the joint positions that is adopted during the initiation of concentric phase of a specific movement has higher correlation to the performance of the movement than isometric force produced at other joint positions.[16,17] For example, performance of snatch and total weight lifted (snatch, clean, and jerk) was observed to have higher correlation with isometric peak force at the starting position than at mid-thigh pull position.[16] Thus, a PAPE activity that aims to increase force generation capability at the starting position may benefit the performance of a weightlifting movement.

The acute increase in force generation capability via PAPE has been adopted to enhance training effect using the contrast training method.[18–22] This training method involves performing exercises of contrasting loads, such as a heavy back squat followed by countermovement jump.[20] While the use of contrast training has been reported to result in both acute and long-term improvement in dynamic performance,[4,19–22] these studies have mainly investigated the effects of pairing heavy resistance exercise with a plyometric exercise. Currently, no study has attempted to pair an isometric and ballistic heavy resistance exercise. As the use of ballistic resistance exercises such as power clean and snatch are commonly used in strength training programs, being able to enhance the performance of these exercises during training may result in better long-term adaptations. Therefore, the purpose of the study was to investigate

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the PAPE effect of MVIC at the starting position on power clean performance over a series of contrast sets. It was hypothesized that a prior MVIC would result in improved power clean performance across all contrast sets.

Methods

Subjects

Eighteen male (age: 31 [3.7] y, body mass: 76.8 [9.1] kg, height: 175.0 [5.2] cm, relative power clean 1RM [one-repetition maximum]: 1.26 [0.18] kg·body mass^{-1}) and 2 female (age: 27.5 [3.5] y, body mass: 53.3.8 [2.0] kg, height: 158.5 [4.9] cm, relative power clean 1RM: 1.19 [0.05] kg·body mass^{-1}) resistance-trained individuals were recruited for this study. Inclusion criteria included: (1) 18–40 years old, (2) has at least 1 year of experience in performing the power clean exercise, (3) performs strength training at least 2 sessions per week for the last 1 year prior to participation in this study, (4) has prior experience in performing 1RM test, and (5) has not sustained any injuries 6 months prior to participating in the study. All participants were brief on the experimental procedure and signed the inform consent form prior to participation. The study commenced after receiving ethical approval from the Institutional Review Board of Singapore Sport Institute.

Design

This study used a repeated-measure approach. Participants were required to attend 1 familiarization and 2 testing sessions. During the familiarization session, participants were briefed on the procedure of the study and the 1RM for power clean was determined. During each testing session, which was separated by at least 48 hours, participants performed 3 sets of 3 repetitions of power clean at 70% of 1RM with or without prior MVIC at the starting position. Participants were randomly assigned to begin each condition using an online application (wheelofnames.com). Lifting velocity was measured to determine the power clean performance.

Methodology

During all sessions, participants performed a standardized warm-up which included 5 minutes of self-paced cycling and 10 repetitions of each body weight dynamic stretch including squat, hip hinge, rear lunge, and push up. Participants then performed 5 repetitions of countermovement jump.

During the familiarization session, participants were given a 2-minute recovery period after the standard warm-up. Following that, participants performed submaximal power clean sets at approximately 30%, 50%, 70%, and 90% of each participant's estimated 1RM. Based on the performance of the previous repetition, the load was increased approximately 2.5 to 5 kg. The participants were provided with 3 minutes of recovery with each successful attempt before attempting again. The heaviest load that the participant lifted for 1 repetition was the participant's 1RM.[23] Participants were provided magnesium chalk and were allowed to use their preferred gripping method when performing the power clean.

During each testing session, participants were given a 2 minutes of passive recovery before performing the specific warm-up for power clean. This included 3 repetitions at 30%, 50%, and 60% 1RM. The power clean was performed using an Olympic bar (20 kg for male and 15 kg for female) and bumper weights (Eleiko). Every repetition began with the weight plates in contact with the ground. Participants were provided 2 minutes of recovery period before commencing on the control condition (CON) or the isometric contrast training condition (ISO). The sequence of each condition was randomized. During the CON, participants performed 3 sets and 3 repetitions of power clean at 70% 1RM. This intensity was selected as a previous study reported that power clean peak power was maximized at 70% 1RM.[24] Each set was separated by 3 minutes of recovery. For the ISO, participants performed 1 set of 3 × 2-second MVIC at the starting position with 1 minute of rest prior to performing each set of power clean. Participants were instructed to adopt the starting position with sufficient pretension only to maintain the posture. They were then instructed to pull as hard and as fast as possible for the MVIC. This procedure was performed for 3 sets with 3 minutes of recovery between sets. To set up for the MVIC, the Olympic barbell was loaded with multiple bumper weight plates till a total weight of 300 kg to prevent participants from lifting the barbell.

Power clean performance was determined by the magnitude of barbell velocity. It has been reported that velocity tracking is a valid and reliable method for monitoring power clean performance.[25] Barbell mean (MV; intraclass correlation coefficient = .97; 95% confidence interval [CI], .94 to .99; % coefficient of variation = 2.3) and peak (PV; intraclass correlation coefficient = .98; 95% CI, .95 to .99; % coefficient of variation = 1.2) velocity were measured using a linear position transducer (GymAware, Kinetic Performance Technology). The average value of each measured variable for each set will be used for further analysis.

Participants were asked to rate their perceived exertion based on the CR-10 RPE scale after completing each session.[26] A rating of 0 was associated with no effort and a rating of 10 was associated with maximal effort. Participants were all familiar with the use of the CR-10 RPE scale as they have been using it regularly during their own training.

Statistical Analysis

Repeated-measures analysis of variance with Bonferroni post hoc analysis was performed for each variable. The P value of ≤ .05 was deemed to indicate statistical significance. All assumptions to run analysis of variance were checked beforehand, including normality and sphericity. Wherever suitable and appropriate, Cohen d was computed: (1) trivial effect size if $d < 0.25$, (2) small effect size $d = 0.25$ to 0.50 and, (3) moderate effect size if $d = 0.50$ to 1.0, (4) large effect size if $d > 1.0$.[27] The smallest worthwhile difference or change for MV and PV was calculated (0.2 multiplied by the between-subject SD).[28]

Results

Group mean data for MV and PV and RPE are displayed in Table 1, with individual participants' results for differences between conditions illustrated in Figure 1. Computed minimum worthwhile change for MV = 0.02 m·s^{-1} and PV = 0.03 m·s^{-1}. A significant time effect was observed for both MV and PV. In the ISO condition, there were significant differences between set 1 and set 3 for both MV ($P = .025$; $d = 0.55$; 95% CI, –0.052 to –0.004 m·s^{-1}) and PV ($P = .049$; $d = 0.42$; 95% CI, –0.056 to –0.008 m·s^{-1}), and between set 2 and set 3 for PV ($P = .028$; $d = 0.53$; 95% CI, –0.065 to –0.004). In the CON condition, there were significant differences between set 1 and set 2 for MV ($P = .012$; $d = 0.62$; 95% CI, –0.069 to –0.009 m·s^{-1}), and between set 1 and set 3 for MV ($P = .009$;

Table 1 Power Clean Mean and Peak Velocity ($m \cdot s^{-1}$) Across 3 Sets

	ISO	Control	Time	Time × group	Group
Mean velocity set 1	1.20 (0.12)	1.14 (0.12)	$F = 11.573$ $P < .001$	$F = 0.821$ $P = .444$	$F = 1.626$ $P = .210$
Mean velocity set 2	1.22 (0.14)	1.18 (0.12)$			
Mean velocity set 3	1.23 (0.14)*	1.18 (0.12)$			
Peak velocity set 1	2.23 (0.14)	2.16 (0.14)	$F = 5.193$ $P = .008$	$F = 0.889$ $P = .415$	$F = 2.075$ $P = .158$
Peak velocity set 2	2.23 (0.14)	2.18 (0.15)			
Peak velocity set 3	2.26 (0.16)*#	2.19 (0.15)			

Abbreviation: ISO, isometric contrast training. Note: Data presented as mean (SD).
*Significantly different from ISO set 1 ($P < .05$). #Significantly different from ISO set 2 ($P < .05$). $Significantly different from control set 1 ($P < .05$).

$d = 0.65$; 95% CI, −0.071 to −0.012 $m \cdot s^{-1}$). No significant difference in PV between sets was observed for CON. Time × group and group effect showed no significant difference for both MV and PV.

The minimum worthwhile change for MV and PV were 0.02 and 0.03 $m \cdot s^{-1}$, respectively. The differences in MV (0.04–0.06 $m \cdot s^{-1}$) and PV (0.05–0.07 $m \cdot s^{-1}$) between the corresponding sets of ISO and CON (eg, ISO set 1 vs CON set 1) resulted in small to moderate effect (MV: $d = 0.38$–0.50, PV: $d = 0.35$–0.50; Table 2). There was no significant difference in RPE between conditions (ISO = 5.0 [1.3], CON = 4.8 [1.4], $P = .385$, $d = 0.22$).

Discussion

The purpose of the study was to investigate the PAPE effect of MVIC at the starting position on power clean performance over a series of contrast sets. The results showed that MV and PV continued to increase from set 1 to set 3 during ISO condition. In addition, differences in MV and PV between the corresponding sets of ISO and CON exceeded the minimum worthwhile change and showed a small to moderate effect in favor of ISO. Hence, in support of the hypothesis, the results suggest that a set of 3 × 2-second MVIC at the starting position with 1-minute recovery period was able to potentiate power clean performance.

Monitoring of lifting velocity may provide insight into an individual's force generation capability and fatigue level.[25,29] In addition, it has been reported that measuring of MV and PV is a valid and reliable method for monitoring power clean performance.[25] The MV is a measure of the average velocity from the start of the first pull until the bar reaches peak height, PV is the highest instantaneous velocity attained during the concentric phase. While coaches may use these values to monitor the power clean performance of their athletes, it is important to also monitor the peak bar height in concurrent with MV as this variable will affect the MV magnitude.

Consistent with previous studies, the current results showed that 3.4% to 5.2% PAPE effect can be induced by performing a set of MVIC prior to a ballistic activity.[4,6–8,10] In addition, the difference in MV and PV of the corresponding sets between both conditions exceeded the minimum worthwhile change in favor of ISO condition. Furthermore, the mean difference in MV and PV also exceeded the typical error (MV = 0.04 $m \cdot s^{-1}$, PV = 0.05 $m \cdot s^{-1}$) observed by Thomas et al[25] when using the same linear transducer to measure MV and PV of power clean at 70% 1RM. This suggests that the difference observed in performance between conditions was likely a meaningful change rather than a random variation in performance. As previous studies have mainly investigated the effect of the PAPE effect of MVIC on jumping performance, the current findings add to the literature that prior MVIC may also induce a potentiating effect for weighted ballistic movements such as the power clean. As the power clean exercise is a commonly used exercise for improving strength and power, and contrast training being shown to be more effective in improving these attributes than traditional strength training,[19,20] the use of MVIC and power clean may now be a viable option for contrast training.

The ISO protocol involved performing MVIC at the starting position of the power clean. This method was proposed as results from various studies showed that the correlation between peak force produced during isometric test had higher correlation to a given dynamic movement when the isometric test was performed at the joint positions that is adopted during the initiation of concentric phase of that specific movement.[16,17] Hence, it is possible that the improved power clean performance in ISO group was attributed to an enhanced force generation capability at the starting position to overcome the initial inertial, leading to a higher rate of acceleration of the barbell. This also suggests that chronic training adaptations that increase maximum force generation at the starting position may enhance power clean performance.

A recovery duration of ≥ 4 minutes is generally required when performing heavy resistance activity to induce PAPE as the fatigue resulting from the heavy resistance activity may negate any potentiating effect if recovery were insufficient.[3,10] However, the current study showed that a shorter duration of 1 minute was sufficient to induce PAPE effect when performing MVIC. This observation was likely due to the shorter recovery time from fatigue induced by performing isometric activity as compared to heavy resistance activity.[5] As isometric activity has been reported to lower levels of muscle soreness, the use of isometric contraction to induce PAPE effect in comparison to heavy resistance exercise may be preferred during periods of competition phase. In further support of the ISO contrast training method, RPE did not differ between conditions, indicating that including the MVIC prior to power clean did not lead to additional perceived fatigue.

Several limitations should be considered while interpreting the results. First, as illustrated in Figure 1, differences in inter-individual response to ISO exist. Although Seitz et al[22] indicated that stronger individuals were able to induce PAPE during contrast sets of heavy and ballistic exercise, the current study was not able to determine the reason for nonresponders as there was no clear difference in strength level (based on relative 1RM power clean). Second, there were only 2 female participants in this study. Hence, the current findings were not able to determine if there were any sex differences in the response to ISO. Previous study by Koźlenia and Domaradzki[8] reported sex difference in the PAPE response over time. The present study observed that

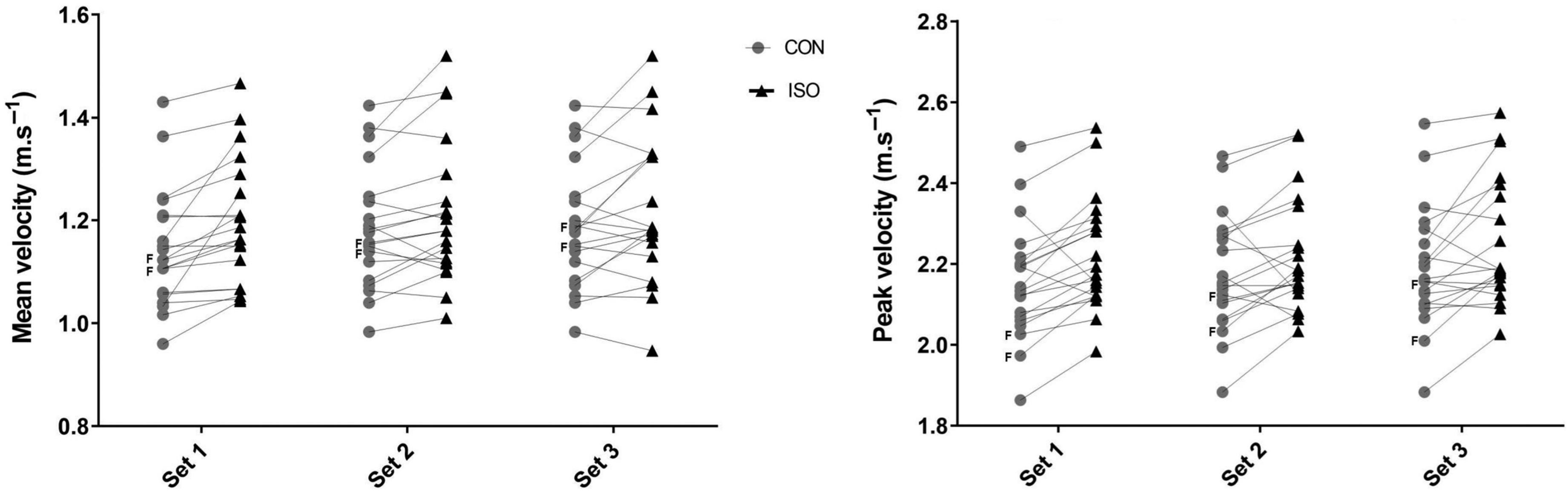

Figure 1 — Intraindividual difference between conditions in power clean mean and peak velocity across 3 sets. CON indicates control condition; ISO, isometric contrast training condition; F, female participant.

Table 2 Between-Conditions Differences in Mean and Peak Velocity

	Difference	*D*
Mean velocity		
ISO vs control set 1	0.06 (0.05)	0.50
ISO vs control set 2	0.04 (0.03)	0.31
ISO vs control set 3	0.05 (0.05)	0.38
Peak velocity		
ISO vs control set 1	0.07 (0.08)	0.50
ISO vs control set 2	0.05 (0.09)	0.34
ISO vs control set 3	0.07 (0.09)	0.45

Abbreviation: ISO, isometric contrast training condition.

both female participants responded to ISO differently. While both showed improved performance during the first set, only one of them showed improvement for the second and third sets. Hence, a study that includes a larger sample size of female participants will be required to determine if sex difference in respond to ISO is present. Finally, power clean performance was only determined by tracking bar velocity. Hence, the mechanisms for the improvement in performance were not determined. Future studies may include force plate measures and video analysis to better understand the mechanisms for enhanced performance.

Practical Applications

Although contrast training has consistently been reported to result in greater positive training adaptations than traditional strength training, the additional sets of exercise may result in longer training time (due to long rest periods) and possibly greater level of fatigue. However, the current results indicate that a short rest period of 1 minute is sufficient when performing MVIC to potentiate subsequent activity. Furthermore, PAPE effect was sustained across 3 sets during the ISO contrast protocol and did not result in increased perceived exertion. Hence, coaches and athletes may consider using MVIC to induce PAPE as this method has been shown to enhance performance for ballistic weighted exercise in this study and jump performance in previous studies.

Conclusion

In conclusion, the present investigation suggests that power-clean performance, as indicated by bar velocity, can be enhanced by a prior maximal voluntary isometric contraction (MVIC) at the starting position with a 1-minute recovery period, as the isometric task likely induced a postactivation performance enhancement (PAPE). Furthermore, there was no increase in perceived effort when a set of MVICs was included prior to the power clean, suggesting that isometric tasks can induce PAPE without causing significant fatigue.

Acknowledgment

The authors would like to thank the participants for their support and participation in this study.

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